

Volve Field NPT Analysis

Quantifying downtime, cost, and its (non-)relationship to production - built on public Equinor Volve data

- Headline
- Pareto
- Wellbore
- Cost
- NPT vs Production

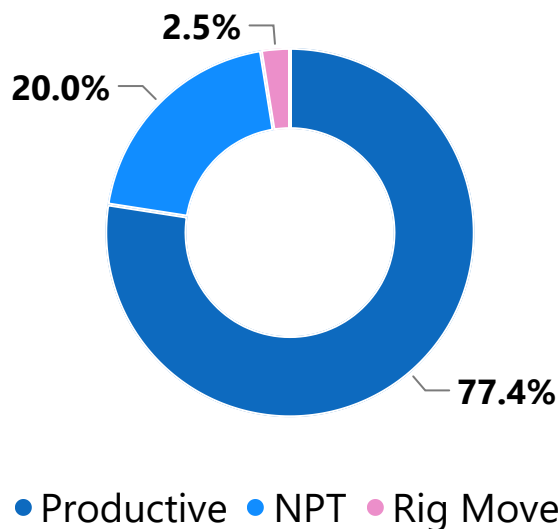
NPT %
20.0%

Total Hours
39,395

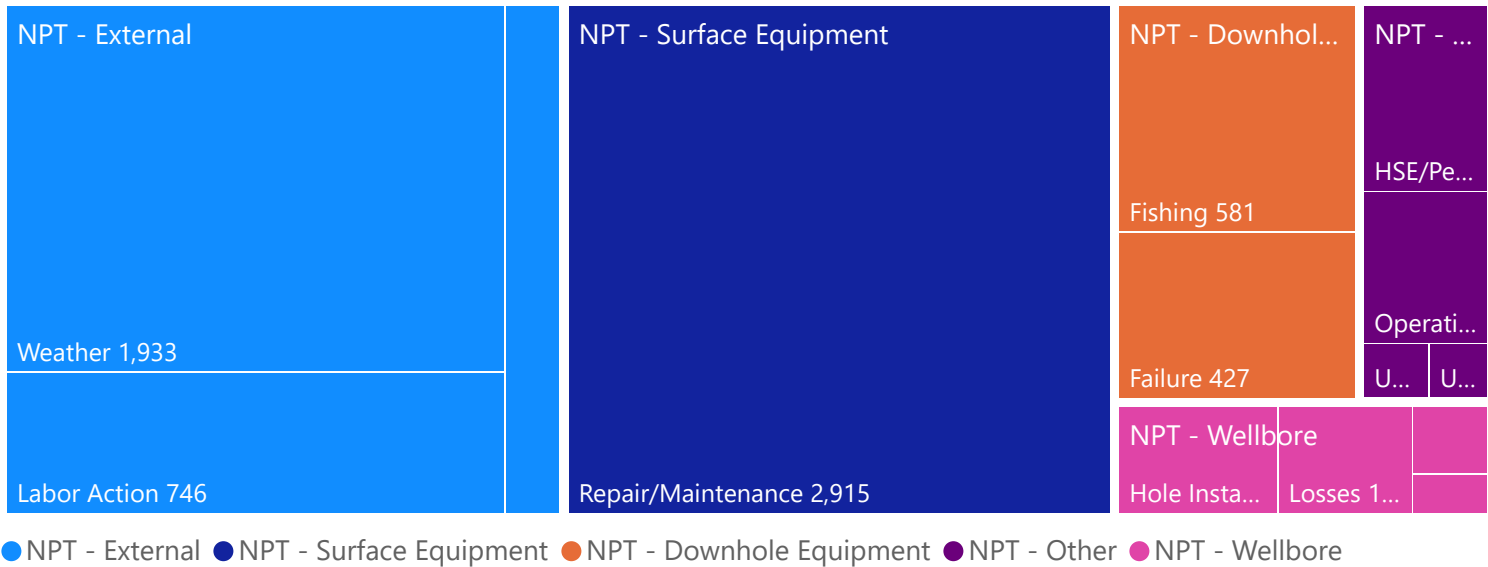
NPT Hours
7,888

Total NPT Cost
\$98.60M

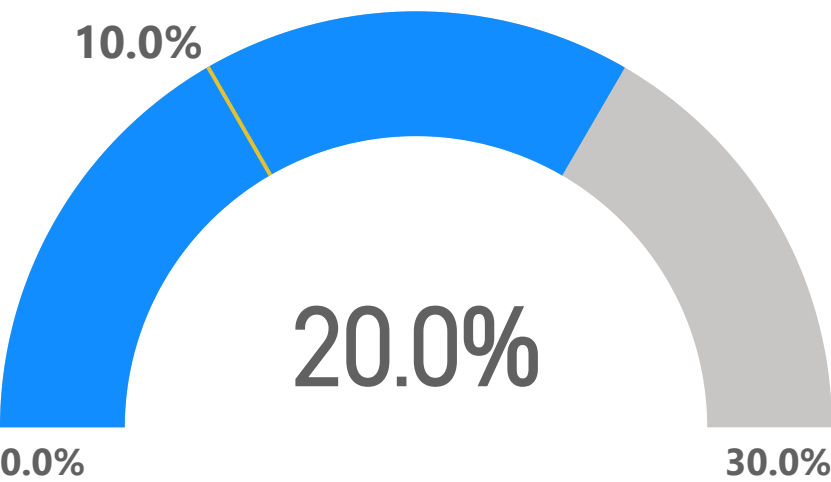
Where the hours went
Hours by Category



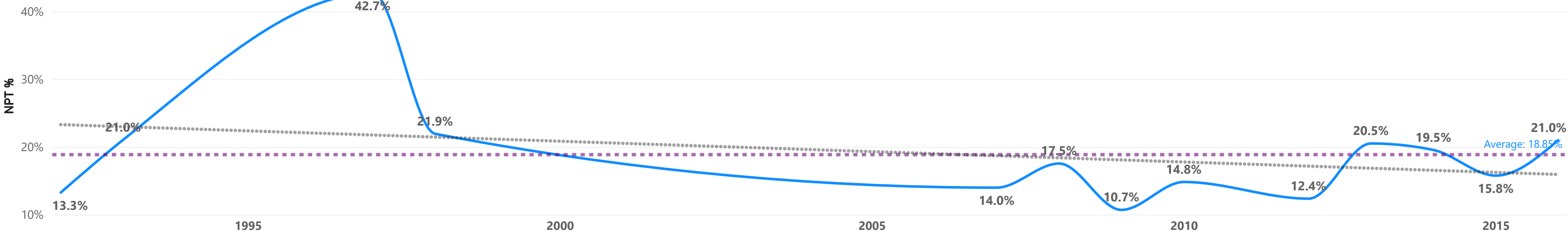
NPT Hours by category (Level & Level 2)



NPT % vs 10% benchmark



Annual NPT %
NPT % by year



Two categories own 62% of all NPT
Surface equipment repair and Weather account for 4,848 of 7,888 NPT hours

Headline

Pareto

Wellbore

Cost

NPT vs Production

NPT Hours

7.89K

Top 1 Share

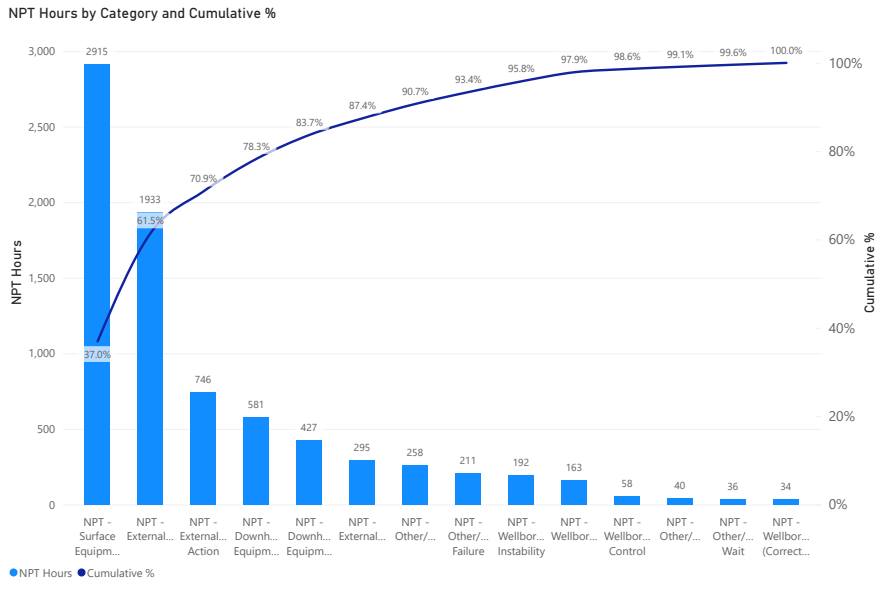
36.95%

Top 2 Share

61.46%

Top 5 Share

83.70%



Category Detail

Category Label	NPT Hours	% of Total NPT
NPT - Surface Equipment/Repair/Maintenance	2,914.86	36.95%
NPT - External/Weather	1,933.25	24.51%
NPT - External/Labor Action	746.00	9.46%
NPT - Downhole Equipment/Fishing	581.33	7.37%
NPT - Downhole Equipment/Failure	426.91	5.41%
NPT - External/Materials/Logistics	295.14	3.74%
NPT - Other/HSE/Permits	258.08	3.27%
NPT - Other/Operational Failure	211.00	2.67%
NPT - Wellbore/Hole Instability	192.17	2.44%
NPT - Wellbore/Losses	162.50	2.06%
NPT - Wellbore/Well Control	57.50	0.73%
NPT - Other/Unclassified	40.00	0.51%
NPT - Other/Unspecified Wait	36.00	0.46%
NPT - Wellbore/Sidetrack (Corrective)	33.50	0.42%
Total	7,888.24	100.00%

Two Short Wells Have the Worst NPT Rates

19-B lost 78.5% of Rig hours; Best producers ran at 10%-18%

Headline

Pareto

Wellbore

Cost

NPT vs Production

Well Type

Development

Exploration

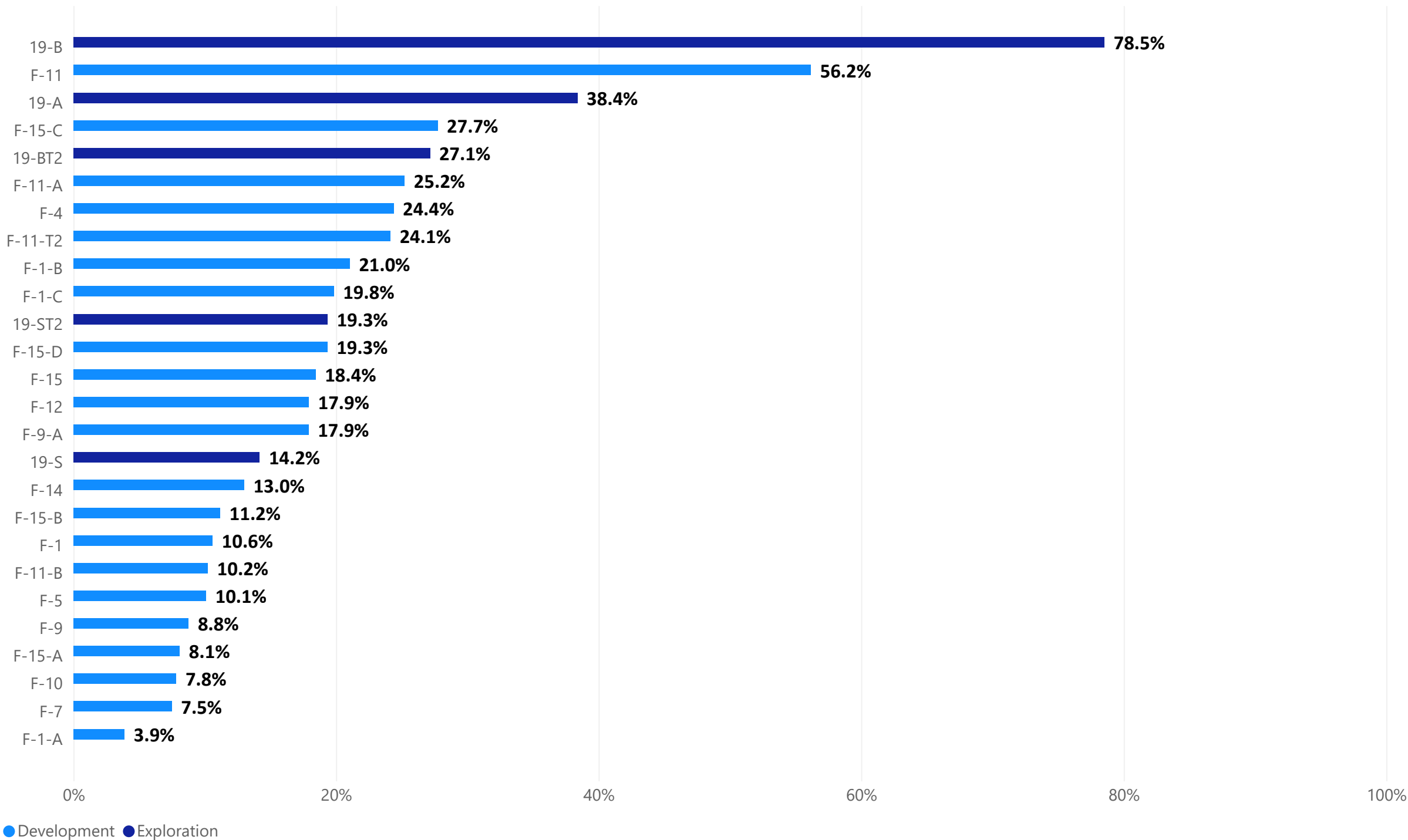
Parent well

All



NPT %, Total Hours, Productive Hours and NPT Hours

by wellbore_key and Well Type



Wellbores

26

Worst NPT %

78.5%

Worst performing W...

19-B

Best NPT %

3.9%

Best performing Well

F-1-A

Concentration

13 of 26 wells cause 82% of NPT

Two Short Wells Have the Worst NPT Rates:

19-B and F-11 lost over half their rig hours, but each logged under 650 hours. Exploration wells average 29.8% NPT against 17.2% for development wells.

NPT cost an estimated \$ 98.6M

5 wells account for about \$43M of modeled downtime cost

Headline

Pareto

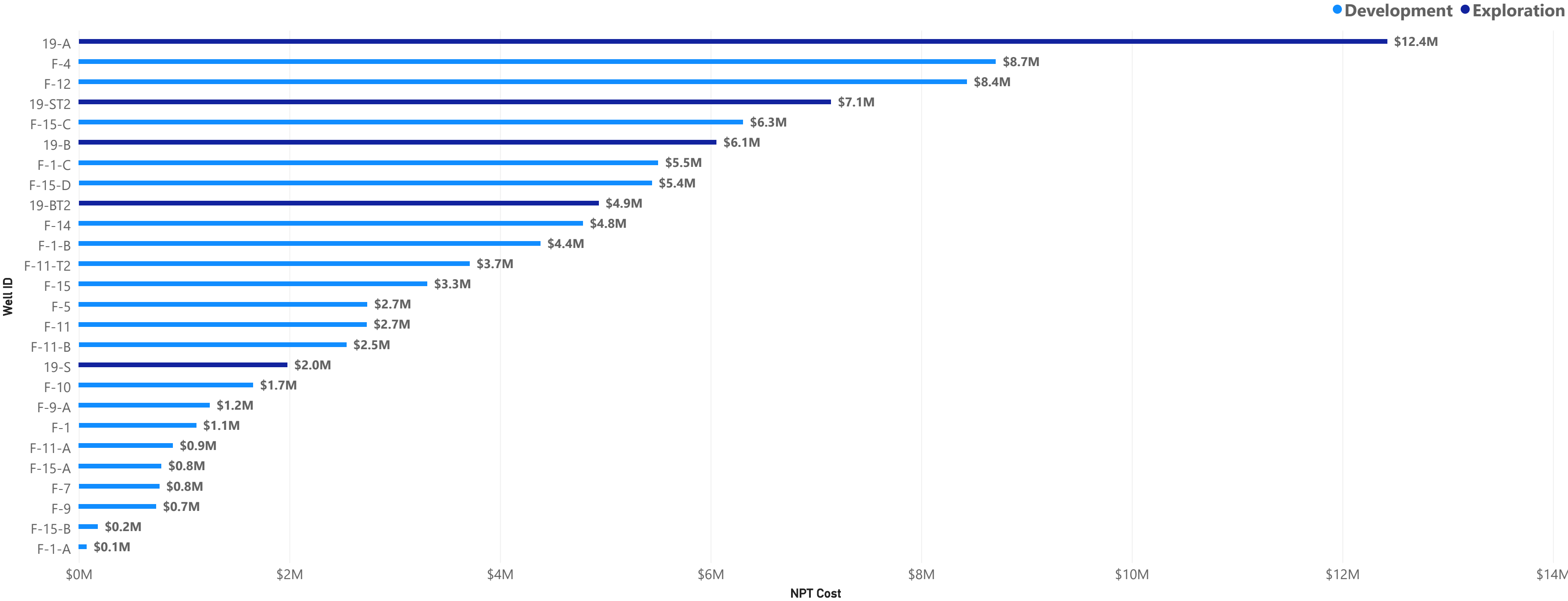
Wellbore

Cost

NPT vs Production

Total NPT Cost	NPT Hours	Assumed Rig Rate	Costliest well	
\$98.60M	7888	\$12,500/hr	\$12.43M	19-A

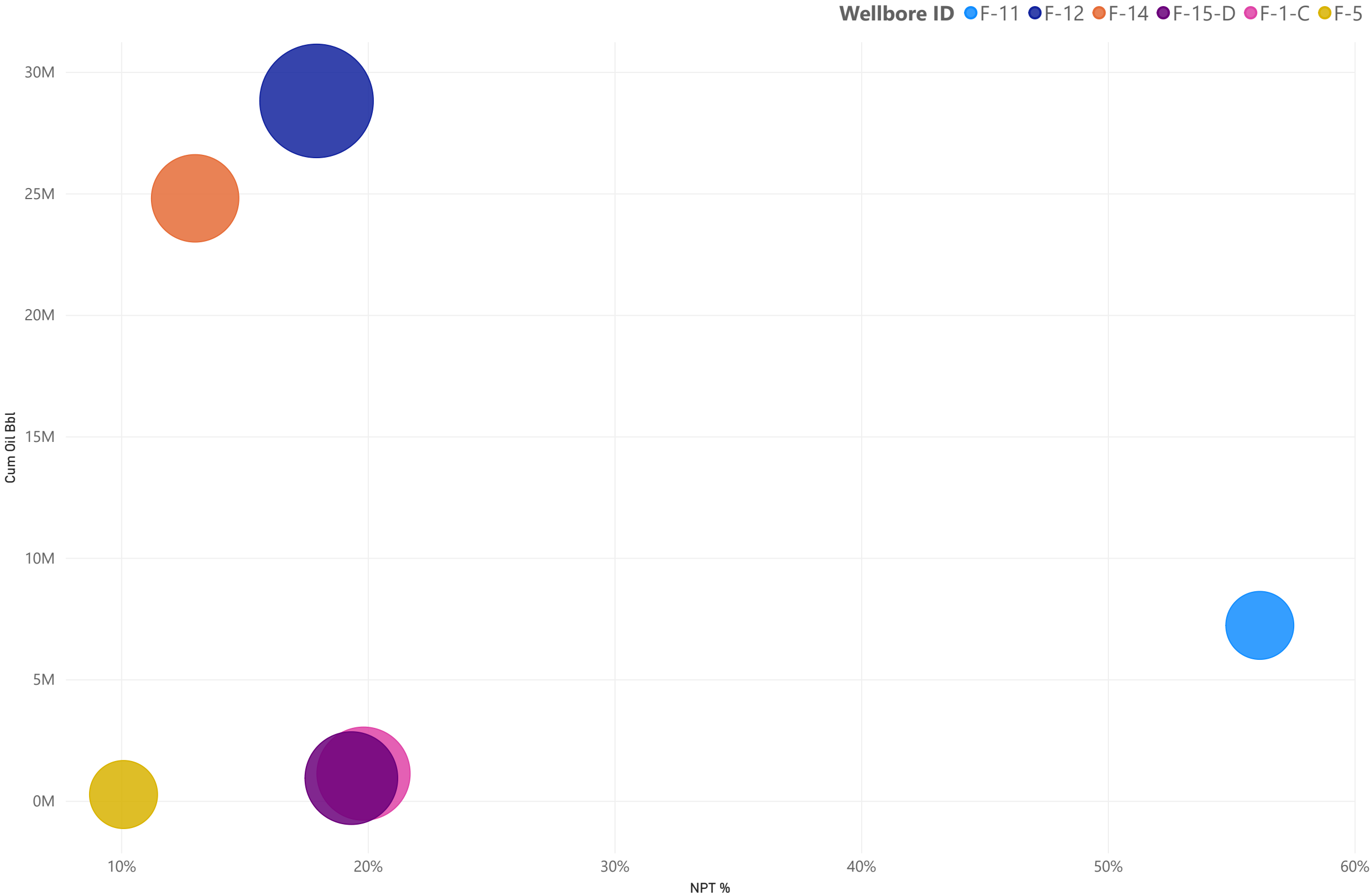
Total NPT Cost and NPT Hours
by wellbore_key and Well Type



High NPT did not predict lower production

Oil producers only ; Bubble size shows modeled NPT cost

NPT %, Cum Oil Bbl and Total NPT Cost
by wellbore_key and wellbore_key



What the Data says:

- F-14 ran at 13.0% NPT and was the most efficient producer, at 0.015 NPT hours per thousand barrels.
- F-12 ran at 17.9% NPT and produced the most oil, 28.8 Mbbl.
- F-11 lost 56.2% of its hours to NPT and still produced 7.2 Mbbl.
- F-5 had 10.1% NPT but only 0.24 Mbbl, the worst hours-per-barrel ratio (0.91).

Conclusion

Among the six wells with oil data, high NPT did not go with low production. The best and the largest producers had moderate NPT, and the highest-NPT well still produced. Output varied by well, and NPT rate doesn't explain it. This is a small sample, so treat it as a pattern and not a proof. Water injectors are excluded.